Part 2 – Workflow and Business Logic Design

1. Workflow Description

To maintain financial control, orders exceeding $5,000 require manager approval before confirmation.

Standard Orders (≤ $5,000)

1. User creates order.
2. System validates items and calculates total.
3. Order is automatically confirmed.
4. Customer receives confirmation notification.

High-Value Orders (> $5,000)

1. User creates order.
2. System calculates total.
3. If total exceeds $5,000:
    - Status is set to **Pending Approval**.
    - Manager is notified.
4. Manager reviews order details.
5. Manager either:
    - Approves → Order is confirmed.
    - Rejects → Order is marked rejected and customer is notified.

This ensures oversight without slowing down standard transactions.

2. Sequence of Steps

1. Order creation and validation
2. Total calculation
3. Threshold check

4. If below threshold → Confirm

5. If above threshold →
    - Set status to Pending Approval
    - Notify manager
    - Await decision
6. Manager approves or rejects
7. System updates status and notifies customer

3. Clarifying Questions

1. Is the $5,000 threshold configurable?
2. Is approval single-level or multi-level?
3. Is there a time limit for approval?
4. Can rejected orders be edited and resubmitted?
5. Should approval actions be logged for auditing?

4. Risks and Challenges

Approval delays – May slow order processing.
Hardcoded threshold – Reduces flexibility.
Notification failure – Manager may not receive alert.
Lack of audit logs – Reduces accountability.